

| Calculations                                                         |            |            |            |            |
|----------------------------------------------------------------------|------------|------------|------------|------------|
|                                                                      | Sheep      | Pig        | Cow        | Goat       |
| Meat Diet Portion                                                    | 2.23%      | 7.21%      | 6.87%      | 1.89%      |
| Meat MCal Target                                                     | 61,082.39  | 197,396.75 | 188,030.12 | 51,715.76  |
| Potential Dairy Value attached to each juvenile                      | 132.54     | 0.00       | 1015.20    | 126.90     |
| Value per juvenile                                                   | 48.00      | 160.00     | 640.00     | 48.00      |
| Adjusted Dairy Value attached to juvenile based on Dairy Utilization | 132.54     | 0.00       | 1015.20    | 126.90     |
| Culled Mothers Value                                                 | 14.77      | 4.41       | 105.26     | 14.77      |
| Culled Adult Males                                                   | 0.74       | 0.22       | 5.26       | 0.74       |
| Final Value per juvenile                                             | 63.51      | 164.63     | 750.53     | 63.51      |
| Value MCals                                                          | 63.51      | 164.63     | 750.53     | 63.51      |
| Offspring required Year <sup>-1</sup>                                | 961.81     | 1,199.06   | 250.53     | 814.32     |
| MCals from meat                                                      | 61082.39   | 197396.75  | 188030.12  | 51715.76   |
| Max Utilized Mcals allowed from Dairy                                | 127478.40  | 0.00       | 254339.08  | 103337.55  |
| MCals actually Utilized from Dairy                                   | 127478.40  | 0.00       | 254339.08  | 103337.55  |
| Percent of diet from their meat                                      | 2.23%      | 7.21%      | 6.87%      | 1.89%      |
| Percent of Diet from their Dairy                                     | 4.66%      | 0.00%      | 9.29%      | 3.77%      |
| Kg Fleece                                                            | 7,268.92   | 0.00       | 0.00       | 0.00       |
| 1 Year+ (age) skins                                                  | 357.68     | 45.87      | 48.28      | 302.87     |
| Yearly Offspring skins                                               | 1,750.41   | 1,413.16   | 369.32     | 1,482.09   |
| Birthing Events per year                                             | 1.00       | 1.86       | 1.00       | 1.00       |
| Offspring Mother <sup>-1</sup> Year <sup>-1</sup>                    | 0.56       | 5.48       | 0.68       | 0.56       |
| Mothers Step 1                                                       | 1,717.52   | 219.01     | 368.43     | 1,454.15   |
| Males Step 1                                                         | 480.91     | 599.53     | 125.27     | 407.16     |
| Females Step 1                                                       | 480.91     | 599.53     | 125.27     | 407.16     |
| Replacement Mothers per year                                         | 343.80     | 44.10      | 46.24      | 291.13     |
| Breeding Males Required                                              | 90.17      | 11.50      | 19.34      | 76.34      |
| Additional Mothers                                                   | 1,408.21   | 39.11      | 174.69     | 1,192.43   |
| Males Step 2                                                         | 394.30     | 107.05     | 59.39      | 333.88     |
| Females 2                                                            | 394.30     | 107.05     | 59.39      | 333.88     |
| Total Mothers                                                        | 2,061.32   | 263.11     | 414.67     | 1,745.28   |
| Total Males                                                          | 875.20     | 706.58     | 184.66     | 741.04     |
| Total Females                                                        | 86.61      | 492.48     | 65.87      | 73.28      |
| Total Offspring                                                      | 1,750.41   | 1,413.16   | 369.32     | 1,482.09   |
| Total Animals                                                        | 3,113.30   | 1,473.66   | 684.54     | 2,635.94   |
| Units per Hectare                                                    | 1.00       |            |            |            |
| Total Adults per Hectare                                             | 1.44       | 0.23       | 0.36       | 1.44       |
| Total Offspring per Hectare                                          | 0.71       | 1.00       | 0.22       | 0.71       |
|                                                                      | 1.53%      |            | 4.32%      | 1.30%      |
| Total Adults                                                         | 3,560      | 314        | 609        | 3,014      |
|                                                                      |            |            |            |            |
|                                                                      |            |            |            |            |
| Total Offspring                                                      | 1,750      | 1,413      | 369        | 1,482      |
| Days for Maintenance                                                 | 5.00       | 220.00     | 0.00       | 5.00       |
| Dry Mothers                                                          | 310.91     | 0.00       | 45.35      | 263.19     |
| Other                                                                | 28,006.54  | 0.00       | 0.00       | 23,713.37  |
| MCals Dry Mothers                                                    | 363,148.13 | 0.00       | 96,006.36  | 307,407.74 |
| total Maintenance MCals                                              | 391,154.67 | 0.00       | 96,006.36  | 331,121.11 |

| MCals Required                  |             |             |             |             |
|---------------------------------|-------------|-------------|-------------|-------------|
|                                 | Sheep       | Pig         | Cow         | Goat        |
| Gestating                       | 1,042,368   | 645,180     | 618,240     | 882,582     |
| Lactating                       | 1,988,639   | 229,356     | 664,774     | 1,683,797   |
| Maintenance                     | 391,155     | 0           | 96,006      | 331,121     |
| Adult Males                     | 105,318     | 0           | 40,948      | 89,168      |
| Offspring                       | 480,487     | 1,420,229   | 1,351,707   | 406,833     |
| TOTAL                           | 4,007,967.8 | 2,294,764.8 | 2,771,674.4 | 3,393,501.4 |
| Animal Diet Breakdown           |             |             |             |             |
|                                 | Sheep       | Pig         | Cow         | Goat        |
| Crop                            | Barley      | Barley      | Barley      | Barley      |
| Grams <sup>-1</sup> Straw/Chaff | 1000.0      | 0.0         | 4956.0      | 1000.0      |
| MCal kg <sup>-1</sup>           |             |             |             |             |
| Grams <sup>-1</sup> Grain       | 750.0       | 750.0       | 1500.0      | 750.0       |
| MCal kg <sup>-1</sup>           |             |             |             |             |
| Days                            | 0.0         | 0.0         | 0.0         | 0.0         |
| Number of Animals               | 1,750.4     | 1,413.2     | 369.3       | 1,482.1     |
| MCals from Straw/Chaff          | 0.0         | 0.0         | 0.0         | 0.0         |
| MCals from Grain                | 0.0         | 0.0         | 0.0         | 0.0         |
| MCals from other feed           | 0.0         | 0.0         | 0.0         | 0.0         |

| Diet Calculations                           |                  |         |
|---------------------------------------------|------------------|---------|
| Plants Mcal                                 | 1,704,844.965    |         |
| Meat Mcal                                   | 547,500          |         |
| Dairy                                       | 485,155.03458812 |         |
|                                             | Perfect          |         |
| Plants %                                    | 62.28%           |         |
| Meat %                                      | 20.00%           |         |
| Dairy %                                     | 17.72%           |         |
| check                                       | Perfect          |         |
| Percent                                     |                  |         |
| Domestic Meat                               | 18.20%           |         |
| Dairy                                       | 17.72%           |         |
| Wild Meat                                   | 1.80%            |         |
| Crops                                       | 56.05%           |         |
| Wild Plants                                 | 6.23%            |         |
| check                                       | Perfect          |         |
|                                             |                  |         |
| new dairy                                   | 485155.03        |         |
| potential dairy < limit                     | TRUE             |         |
| new limit                                   | 80.0%            |         |
| the limit greater than that left to satisfy | TRUE             |         |
| limit                                       | 1                |         |
| limit dairy checked                         | FALSE            |         |
| Mcalories                                   |                  | Surplus |
| Domestic Meat                               | 498,225.000      |         |
| Dairy                                       | 485,155.035      |         |
| Wild Meat                                   | 49275.000        |         |
| Crops                                       | 1,534,360.469    |         |
| Wild Plants                                 | 170,484.497      |         |
| check                                       | Perfect          |         |
|                                             |                  |         |
| SURPLUS DAIRY                               | 0.000            |         |
| This could account for additional % of diet | 0.00%            |         |

Herd Modelling

|                                     | Sheep   | Pig     | Cow     | Goat    |  |  |  |  |
|-------------------------------------|---------|---------|---------|---------|--|--|--|--|
| Mothers Req'd                       | 2061    | 263     | 415     | 1745    |  |  |  |  |
| Males for Slaughter                 | 875     | 707     | 185     | 741     |  |  |  |  |
| Females for Slaughter               | 87      | 492     | 66      | 73      |  |  |  |  |
| Offspring available for Consumption | 1750    | 1413    | 369     | 1482    |  |  |  |  |
| Gestation Req'd                     | 572758  | 547431  | 419389  | 484929  |  |  |  |  |
| Lactation Req'd                     | 1092713 | 194607  | 450956  | 925152  |  |  |  |  |
| Maintenance Req'd                   | 1299619 | 0       | 123706  | 1100384 |  |  |  |  |
| Adult Males                         | 105318  | 0       | 40948   | 89168   |  |  |  |  |
| Offspring Required                  | 575009  | 1547413 | 1617616 | 486865  |  |  |  |  |
| Energy Required                     | 3645419 | 2289451 | 2652615 | 3086499 |  |  |  |  |
| Ewe Fleece Kg Yr-1                  | 4638    |         |         |         |  |  |  |  |
| Ram Fleece Kg year-1                | 180     |         |         |         |  |  |  |  |
| Juvenile Fleece Kg Yr-1             | 2451    |         |         |         |  |  |  |  |
| Kg Fleece Year-1                    | 7269    | 0       | 0       | 0       |  |  |  |  |